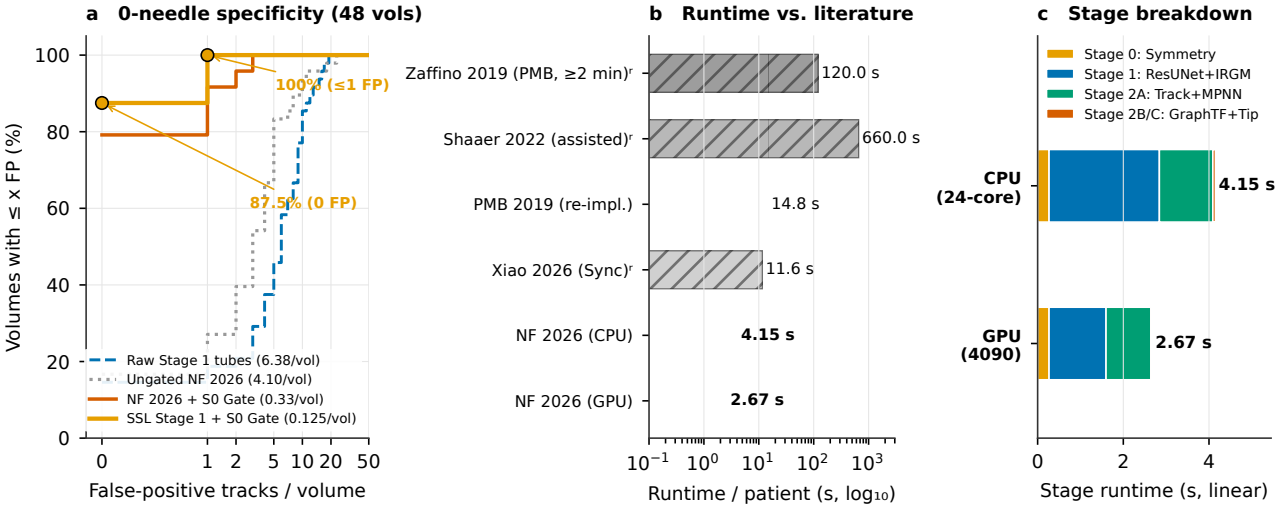
a 0-needle specificity (48 vols)
Volumes with ≤ x FP (%)
False-positive tracks / volume
100 %
87.5%
100 % (≤1 FP)
87.5% (0 FP)
Raw Stage 1 tubes (6.38/vol)
Ungated NF 2026 (4.10/vol)
NF 2026 + S0 Gate (0.33/vol)
SSL Stage 1 + S0 Gate (0.125/vol)
b Runtime vs. literature
Zaffino 2019 (PMB, ≥2 min)r
Shaaer 2022 (assisted)r
PMB 2019 (re-impl.)
Xiao 2026 (Sync)r
NF 2026 (CPU)
NF 2026 (GPU)
120.0 s
660.0 s
14.8 s
11.6 s
4.15 s
2.67 s
Runtime / patient (s, log10)
c Stage breakdown
Stage 0: Symmetry
Stage 1: ResUNet+IRGM
Stage 2A: Track+MPNN
Stage 2B/C: GraphTF+Tip
CPU (24-core)
GPU (4090)
4.15 s
2.67 s
Stage runtime (s, linear)